Biodiversity in the 21st Century at Gbg University

Tomek Garbus

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1 Reading 1: Diaz et al 2019

1.1 Summary

- **telecoupling** – human actions incresingly act at a distance due to globalization
- increasing demand between supply and demand due to global trade
- the analysis pinpoints five crucial levers (priority interventions) and eight leverage points (for intervention)

1.2 Intro

- the human impact on life on Earth has increased since the 1970s
- both the benefits of economy and the cost of reducing nature are unequally distributed

1.3 Taking stock of the fabric of life

- over the past 50 years, the quality of nature to support life has declined on 14 of 18 categories identified by IPBES [1]
- exceptions to the downward trend are: regulation of ocean acidification, energy, food and feed, materials and assistance
- more than 800 million people still face chronic food deprivation
- the biomass of world's vegetation has halved over human history
- forest area is only 68% of its preindustrial size

1.4 Direct and indirect drivers of change

- Direct drivers:
  - land/sea use change
  - direct exploitation
  - climate change
  - pollution
  - invasive alien species
  - others
- Indirect drivers:
  - demographic and sociocultural
  - economic and technological
  - institutions and governance
  - conflicts and epidemics
- examples of declines in nature:
  - natural ecosystems have declined by 47% on average, relative to their previous states
  - 25% of species are already threatened by extinction
  - biotic integrity – abundance of naturally present species – has declined by 23% on average in terrestial communities
  - the global biomass of wild mammals has fallen by 82%
  - 72% of indicators developed by Indigenous Peoples and local communities have deteriorated

1.5 Progress towards internationally agreed goals

- 20 Aichi Targets in the Strategic Plan on Biodiversity 2011-2020
  - contains 54 elements
  - good progress has been made on 5
  - moderate progress towards 19
  - poor progress or movement away from target on 21
  - unknown progress on 17 elements
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1.6 Levers and leverage points for transformative change

Levers:

- incentives and capacity building
- cross-sectional cooperation
- pre-emptive action
- decision-making in the context of resilience and uncertainty
- environmental law and implementation

Leverage points:

- embrace diverse vision of a good life
- reduce total consumption and waste
- unleash values and action
- reduce inequalities

[1] Intergovernmental Platform on Biodiversity and Ecosystem Services

- practice justice and inclusion in conservation
- internalize externalities and telecouplings
- ensure environmentally friendly technology, innovation and investment
- promote education and knowledge generation and sharing

2 Lecture 1

2.1 Housekeeping

- Every Tuesday
- 18 - ca. 20h
- Live or recorded on Zoom
- All lectures will be recorded and available on Canvas
- Two in-person meetings
- Gothenurg Natural History Museum
- Gothenburg Botanical Garden
- Assignment: develop and submit an essay on a topic we cover in the course that is of particular personal interest. Open-ended and offers the opportunity for personal reflection, while also gaining experience in more scientific-styled writing.

2.2 Biodiversity bias

- 2 million – 1 trillion species on Earth
- ca. 1.7 million databased
- Knowledge derives from charismatic, macroscopic species.

2.3 Mechanisms of biodiversity formation

- speciation
- extinction
- migration

Adaptation = the process which enables organisms to adjust to their environment in order to ensure survival. This can lead to speciation.

Example: polar bears have extremely large feet to distribute weight better on snow.

2.4 Speciation

Types of speciation:

- allopatric – populations get geographically distanced which limites the gene flow
- peripatric – a small group gets isolated
- parapatric – populations are adjacent but not completely sepatated
- sympatric – species evolve from a single ancestral species while living in the same area

2.5 Extinction

The termination of any organism.

Species have a lifespan, on average 2 million years. After that, species either speciate or go extinct.

2.6 Dispersal

Dispersal = movement of individuals (through geological time, such as marsupials colonizing North America once South and North Americas got connected through geological time)

Since the industrial revolution, many species are progressively changing their distribution and dispersing further north.

2.7 Latitudinal species gradient

Closer to equator/tropics = more species (also more human languages!)

Does the diversity in species and language covary?

2.8 Species

How do we measure species for example in tropical jungle with countless species?

Quandrants: count within a small area then you can extrapolate

Species richness = the number of different species in an area

Numerical species richness = number of species per specific number of individuals

Species density = number of species per unit area

Species abundance = the relative abundance of species

Amazonia is known to be highly biodiverse, but a few species are very abundant and most species are incredibly rare. 1% of species make up for 50% of all trees.

2.9 Learning outcomes

- Biodiversity: number of differential biological variants found in a given place and time
- Speciation, extinction, and dispersal build biodiversity
- Biodiversity is not evenly distributed and can be measured in different ways

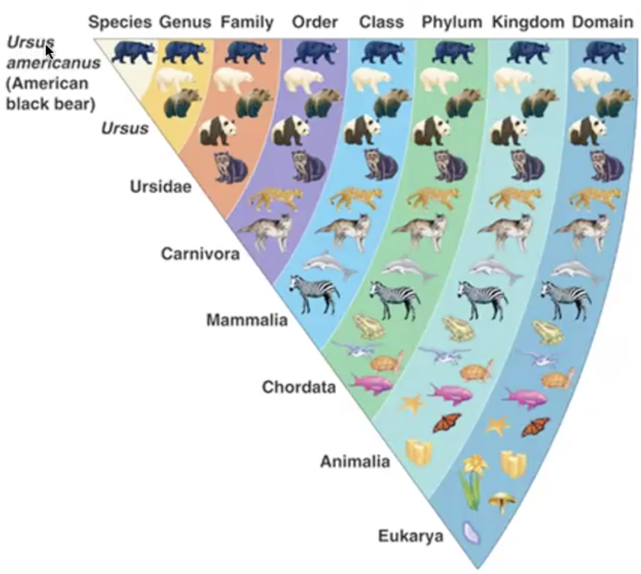
3 Reading 2: Forest et al 2018: Gymnosperms on the EDGE

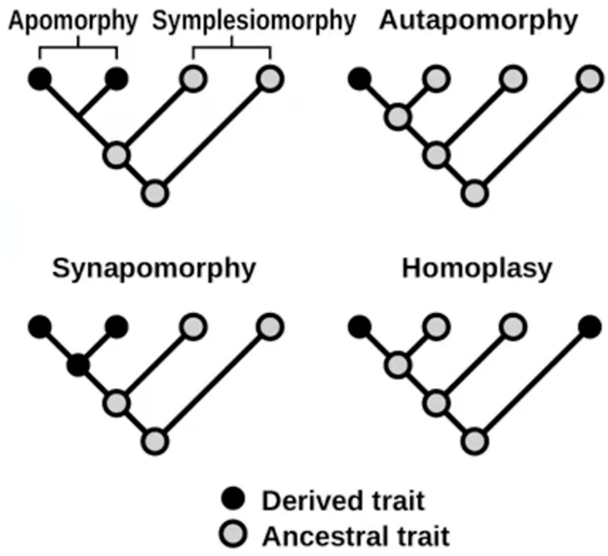
3.1 Gymnosperms

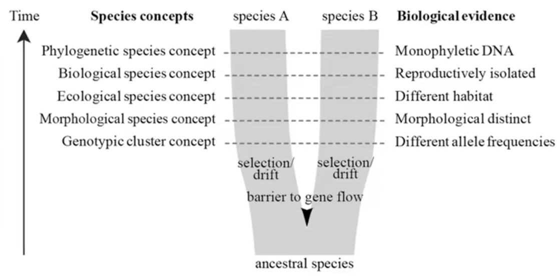
- Comprising ginkgo, conifers, cycads, and gnetophytes
- 40% species at high risk of extinction
- For plants in general, this number is 20%

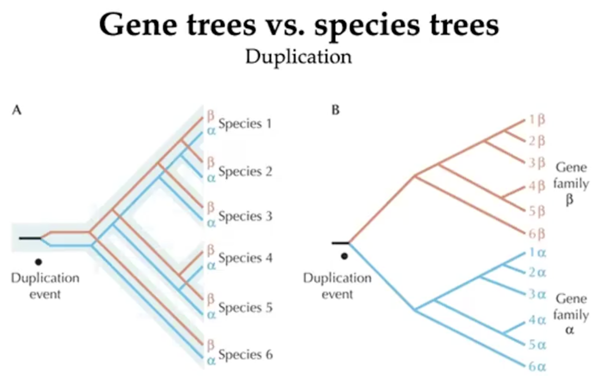
3.2 Linneus' binomial classification rules

1. The entire 2-part name must be written in italics.
2. The genus name must be written first.
3. The genus name must be capitalized.
4. The specific epithet is never capitalized.









4 Climate, Oxygen, and the Future of Marine Biodiversity

- The ocean enabled the diversification of life on Earth by adding O_2 to the atmosphere.
- Human industrialization is intensifying the aerobic challenges to marine ecosystems.
- Historical observations report an 2% decline in upper-ocean O_2.
- The ocean retains under 2% of global O_2 inventory, while 98% stays in the atmosphere.

- The constraining role of O_2 for ocean species arises from two kinematic factors: lower solubility of gases in water and slower circulation of the ocean than of atmosphere.
- Industrialization of human societies produced two negative consequences for the ocean's O_2 reservoir: (1) nutrients mobilized for agriculture are carried to coastal oceans, stimulating phyloplankton growth and thus higher O_2 respiratory demand (2) global warming causes O_2 depletion throughout the ocean
- Coastal regions make up only 7% of ocean's area but contain 15% of it's net primary productivity and 90% of it's animal life.
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4.1 Glossary

4.1.1 O_2 solubility in seawater (K_H)

O_2 in seawater when in equilibrium with the O_2 in a gas phase, as a function of temperature, pressure, and salinity

4.1.2 O_2 partial pressure (pO_2)

the ratio of O_2 concentration to O_2 solubility

4.1.3 hypoxia

the condition wherein O_2 is insufficient to meet metabolic demand, which varies with species and temperature, in contrat to fixed definition commonly used in ocanography (O_2 ¡ 60μM)

4.1.4 Eutrophication

increased growth, primary production, and biomass of algae in response to nutrient enrichment in the environment (e.g. sewage, fertilizer, and detergents)

4.1.5 Metabolic storm

an anomalous ocean weather event in which high temperatures cause elevated organismal O_2 demand simultaneous with decline in O_2 available in seawater

4.1.6 Temperature-dependent hypoxia

the threshold for hypoxia and its variation with temperature

4.1.7 Oxygen minimum zone (OMZ)

a vertical minimum in a water column; OMZs are globally prevalent but are strongest in the tropical thermoclines of the Indian and Pacific basins, where O_2 reaches traditionally defined hypoxic concentrations